

Predictive maintenance of bearing shells

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Chair of Cyber-Physical-Systems

Motivation

- Journal bearing failures cause high maintenance costs and downtime.
- Friction behavior is a key indicator of bearing condition, but it is difficult to model due to noisy and interacting sensor signals.
- Machine learning can help capture these relationships.

Research Question

Can machine learning models predict the coefficient of friction from sensor data, and which model performs better?

Data and Problem

Dataset - Journal Bearing Adapter experiments.

Measurements: torque, load, speed, temperature, and acoustic emission.

Task - regression problem with the coefficient of friction as the target variable.

Method

The data was preprocessed and normalized, and physically motivated features were engineered. Two regression models were trained and compared:

- Random Forest
- MLP
- Model performance was evaluated using RMSE, MAE, and R^2 .

Results

Random Forest model - very high accuracy on the test set ($R^2 \approx 0.99997$).

MLP model - poor generalization with negative R^2 .

Feature importance analysis showed torque as the dominant predictor, which matches physical expectations.

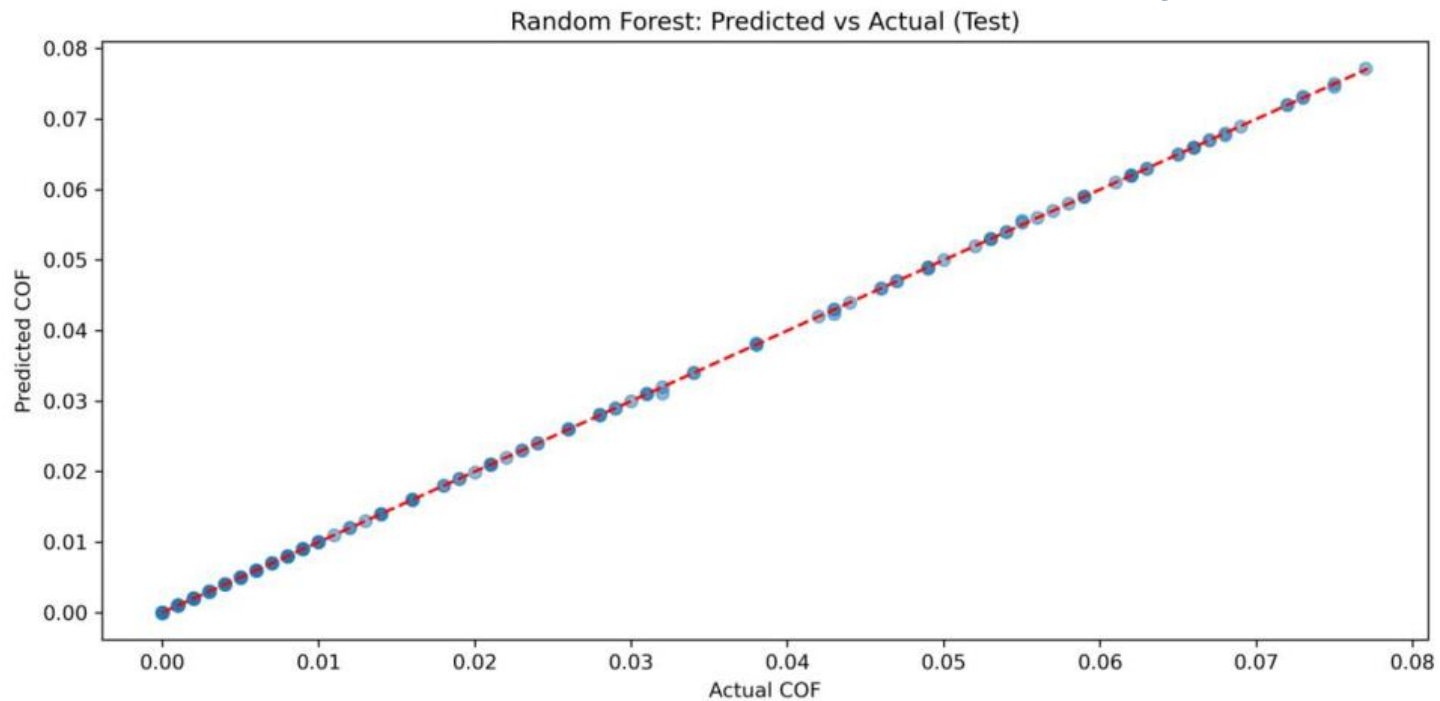


Random Forest clearly outperforms MLP on all metrics.

Model	RMSE (Test)	MAE (Test)	R^2 (Test)
Random Forest	1.20×10^{-4}	3.17×10^{-5}	0.99997
MLP Regressor	4.82×10^{-2}	1.49×10^{-2}	-4.07

Model performance on test dataset

The predictions
match the
actual values,
indicating very
high accuracy.



Random Forest: Predicted vs Actual COF

Conclusion

Random Forest is well suited for friction prediction in this dataset. Direct bearing failure prediction was not possible due to limited wear variability, but accurate friction prediction is a meaningful step toward predictive maintenance.

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Thank You for the Attention!

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